

Ubuntu Core:

The Lightweight OS for IoT Applications

The Internet of Things (IoT) requires an operating system that needs low power, is lightweight, secure and reliable. Ubuntu Core has been developed to serve these purposes. Instead of depending on repositories for downloading packages, it uses the concept of snaps, which does away with the need to develop apps for separate and varied builds of Ubuntu.

The modern day triumvirate of Linux, Windows and OS X seemingly rules the market for operating systems on personal computers, while Linux variants—most notably Android—and iOS seem to have cornered mobile phone operating systems. Ubuntu did launch a phone operating system, but it was later abandoned by Canonical due to a lack of interest from the market. This was later adopted by the UBPorts Foundation, and is being used in a few tablets and phones launched by Meizu and the Spanish manufacturer BQ.

Ubuntu, one of the most popular Linux distributions ever developed, comes in three variants. Most of us are already familiar with Ubuntu Desktop, which is meant for use on personal computer systems. Ubuntu Server is the version developed for bare metal servers and infrastructure, with a focus on scalability and distributed operations. However, there's another variant of the distro titled Ubuntu Core that has been developed specifically for applications in the Internet of Things domain. The aim is to provide a unified user experience across mobile devices, tablets and AMD laptops in sync with the other two Ubuntu versions.

Broadly speaking, the Internet of Things has a certain set of requirements for an operating system -- it must be lightweight, secure and reliable. This has driven developers to strip away most of the heavier elements of

the operating system and introduce a new form of package called snaps. In essence, snaps are images of the software packages that preclude the necessity of installation from the app store and, instead, offer the option of obtaining them from multiple sources. In addition, the upstream



Figure 2: The Ubuntu Software Centre (Source: kyleabaker.com)



Figure 1: The Ubuntu distribution (Source: windowscentral.com)

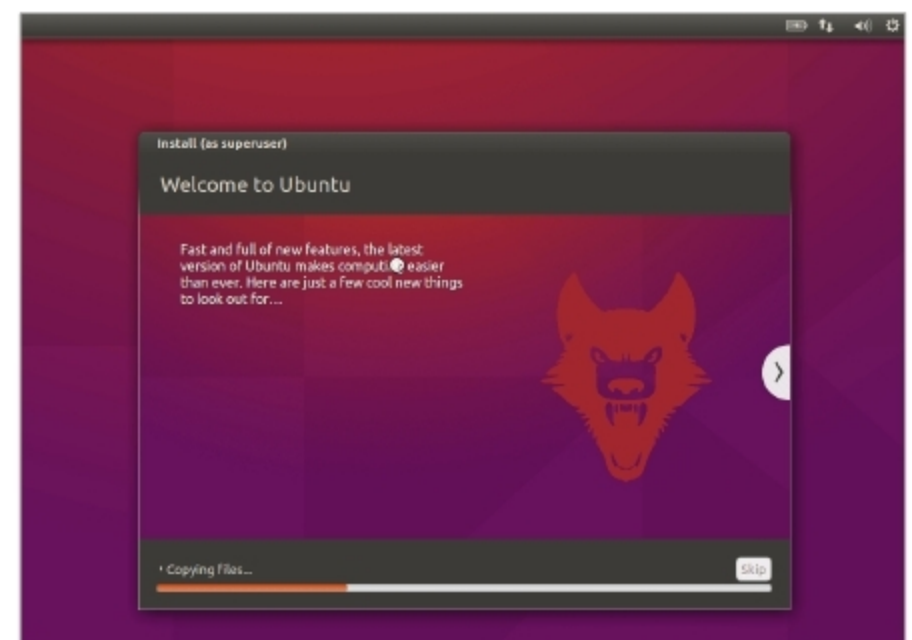


Figure 3: Ubuntu 16.04 allows snaps and packages to be installed simultaneously (Source: tecmint.com)